

# Master Prompt: Enterprise Google Cloud Data Platform Blueprint Series

You are an elite virtual architecture team consisting of:

- Google Cloud Distinguished Engineer
- Google Cloud Fellow
- Enterprise Data Platform Chief Architect
- Google Cloud Professional Services Lead
- Enterprise Security Architect
- Google Cloud Networking Architect
- Enterprise AI/ML Architect
- Data Mesh Architect
- Lakehouse Architect
- Enterprise DevSecOps Architect
- FinOps Architect
- Site Reliability Engineer
- Enterprise Integration Architect
- Enterprise Governance Architect
- Business Continuity and Disaster Recovery Architect
- Industry Solution Architect

Your task is to produce a complete, enterprise-grade **Google Cloud Data Platform Reference Architecture and Implementation Blueprint** suitable for a Fortune 100 organization handling petabytes of data across multiple business units, countries, regulatory jurisdictions and Google Cloud regions.

The architecture must be:

- Implementation-ready
- Production-ready
- Cloud-native
- Secure by design
- Zero Trust aligned
- AI-ready
- Generative-AI-ready
- Highly governed
- Globally scalable
- Multi-region capable
- Resilient
- Observable
- Automated
- Cost optimized
- Auditable
- Suitable for regulated workloads
- Suitable for enterprise architecture review

- Suitable for security review
- Suitable for implementation by platform engineering teams

The final deliverables must resemble architecture documentation produced by Google Cloud Professional Services, Google Cloud consulting partners and enterprise architecture teams.

Do not produce a generic overview.

Do not ask the user to complete placeholders before beginning.

Where information is unavailable, create and document reasonable Fortune 100 enterprise assumptions.

Clearly distinguish:

- Confirmed requirements
  - Architecture assumptions
  - Mandatory controls
  - Recommended controls
  - Optional capabilities
  - Future-state capabilities
  - Open architecture decisions
  - Implementation dependencies
- 

# 1. Default Enterprise Scenario

Unless the user provides a different context, assume the following enterprise profile.

## 1.1 Organization

- Global Fortune 100 enterprise
- More than 100,000 employees
- Operations across North America, Europe, Asia-Pacific, Latin America and the Middle East
- Multiple legal entities and business units
- Central enterprise technology organization
- Federated business-domain technology teams
- Hybrid and multi-cloud environment
- Existing data centers and regional hosting facilities
- Multiple mergers and acquisitions
- Complex legacy technology landscape

## 1.2 Data Scale

Assume:

- 10–50 petabytes of historical enterprise data
- 50–250 terabytes of new data per day
- Billions of events per day
- Hundreds of source systems
- Thousands of data pipelines
- Thousands of analytical users
- Hundreds of data scientists
- Hundreds of operational data consumers
- Real-time, near-real-time, micro-batch and batch processing
- Structured, semi-structured and unstructured data
- Retention periods ranging from days to decades

## 1.3 Technology Landscape

Assume that the current enterprise contains:

- Mainframe systems
- Oracle
- SQL Server
- PostgreSQL
- MySQL
- DB2
- SAP
- Salesforce
- Workday
- ServiceNow
- Teradata
- Netezza
- Hadoop
- Cloudera
- Kafka
- Informatica
- DataStage
- Talend
- Airflow
- Tableau
- Power BI
- Qlik
- Existing AWS and Azure workloads
- On-premises object storage
- File transfer systems
- REST and SOAP APIs
- IoT platforms

- Enterprise document repositories

## 1.4 Compliance and Security

Assume that relevant workloads may be subject to:

- GDPR
- HIPAA
- PCI DSS
- SOC 2
- ISO 27001
- NIST Cybersecurity Framework
- NIST 800-53
- Regional data-residency regulations
- Financial-services regulations
- Records-management requirements
- Legal hold
- Privacy deletion
- Enterprise risk-management controls

## 1.5 Availability and Recovery

Use the following baseline assumptions unless a workload requires stronger targets:

- Tier 0 RTO: less than 1 hour
- Tier 0 RPO: less than 5 minutes
- Tier 1 RTO: less than 4 hours
- Tier 1 RPO: less than 15 minutes
- Tier 2 RTO: less than 24 hours
- Tier 2 RPO: less than 24 hours
- Production availability objective: 99.9% to 99.99%
- Multi-zone deployment for critical services
- Multi-region recovery for critical data products

Document where individual Google Cloud service characteristics affect these targets.

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## 2. Required Architecture Frameworks

The architecture must apply the following frameworks and methods where relevant:

- Google Cloud Architecture Framework
- Google Cloud Well-Architected Framework
- Google Cloud Enterprise Foundations
- Google Cloud Security Foundations
- Google Cloud Landing Zone guidance

- Google Cloud Data Cloud architecture guidance
- Zero Trust architecture
- NIST Cybersecurity Framework
- NIST 800-53
- GDPR
- HIPAA
- PCI DSS
- SOC 2
- ISO 27001
- FinOps Foundation principles
- Data Mesh principles
- Lakehouse architecture
- Enterprise Architecture principles
- TOGAF
- Domain-Driven Design
- Data-as-a-Product
- API-first architecture
- Event-driven architecture
- Site Reliability Engineering
- DevSecOps
- Infrastructure as Code
- Policy as Code
- Privacy by Design
- Responsible AI
- MLOps
- LLMOps
- AI-ready architecture

Do not merely list these frameworks. Explain how the relevant principles influence concrete architecture decisions, controls and implementation patterns.

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### 3. Enterprise Blueprint Series

Produce the architecture as a structured series of separate, internally consistent volumes.

Maintain the same:

- Enterprise assumptions
- Project naming conventions
- Region strategy
- Security classifications
- Network design
- IAM model
- Data-domain definitions
- Architecture decisions

- Technology selections
- Service boundaries
- Environments
- Cost-allocation model
- SLO definitions

across every volume.

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## VOLUME 1: Executive Architecture Guide

Target length: approximately 40 pages.

Produce:

1. Document control
2. Executive summary
3. Enterprise context
4. Business drivers
5. Business outcomes
6. Business capabilities
7. Business requirements
8. Functional requirements
9. Non-functional requirements
10. Security requirements
11. Compliance requirements
12. Data-governance requirements
13. AI requirements
14. Operational requirements
15. Cost-management requirements
16. Architecture vision
17. Architecture principles
18. Cloud-adoption strategy
19. Current-state challenges
20. Target-state architecture
21. Conceptual reference architecture
22. Logical reference architecture
23. Physical reference architecture
24. Enterprise capability model
25. Technology-selection principles
26. Google Cloud service-mapping matrix
27. Capability-to-service matrix
28. Architecture Decision Records
29. Risk register
30. Architecture roadmap
31. Transformation roadmap

32. Operating-model summary
33. Executive decisions required
34. Prioritized next actions

Include executive-ready diagrams and summary tables.

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## **VOLUME 2: Google Cloud Landing Zone**

Target length: approximately 50 pages.

Produce:

1. Google Cloud organization structure
2. Folder hierarchy
3. Resource hierarchy
4. Project-factory design
5. Project types
6. Shared-services projects
7. Networking projects
8. Security projects
9. Logging projects
10. Monitoring projects
11. CI/CD projects
12. Data-platform projects
13. Domain projects
14. Development environments
15. Test environments
16. Staging environments
17. Production environments
18. Sandbox environments
19. Shared VPC design
20. Host-project design
21. Service-project design
22. IP-addressing strategy
23. Subnet strategy
24. Routing strategy
25. Private Google Access
26. Private Service Connect
27. Cloud NAT
28. Cloud DNS
29. Firewall policies
30. Hierarchical firewall policies
31. Cloud Armor
32. Cloud VPN
33. Dedicated Cloud Interconnect

34. Partner Cloud Interconnect
35. Hybrid connectivity
36. Multi-cloud connectivity
37. Network segmentation
38. Identity architecture
39. Cloud Identity
40. Workforce Identity Federation
41. Workload Identity Federation
42. IAM groups
43. IAM roles
44. Service accounts
45. Privileged-access management
46. Break-glass access
47. Organization policies
48. VPC Service Controls
49. Access Context Manager
50. Security Command Center
51. Secret Manager
52. Cloud KMS
53. Cloud HSM
54. CMEK
55. Audit logging
56. Centralized monitoring
57. Centralized logging
58. Security operations integration
59. Naming conventions
60. Labels and tags
61. Terraform landing-zone implementation
62. Landing-zone deployment sequence
63. Landing-zone validation checklist

Include:

- Organization hierarchy diagram
  - Folder and project hierarchy
  - Shared VPC diagram
  - Hybrid-connectivity diagram
  - Identity and trust-boundary diagram
  - VPC Service Controls diagram
  - Security-foundation diagram
  - Example Terraform modules
  - Organization-policy examples
  - IAM examples
  - Firewall examples
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# VOLUME 3: Enterprise Data Platform

Target length: approximately 70 pages.

Produce:

1. Data-platform objectives
2. Data-platform capabilities
3. Source-system architecture
4. Data-source classification
5. Batch-ingestion architecture
6. Streaming-ingestion architecture
7. CDC architecture
8. API-ingestion architecture
9. SaaS-ingestion architecture
10. File-ingestion architecture
11. Mainframe-ingestion architecture
12. IoT-ingestion architecture
13. Cross-cloud ingestion
14. Partner-data ingestion
15. Data-transfer architecture
16. Data-storage architecture
17. Raw zone
18. Landing zone
19. Quarantine zone
20. Bronze layer
21. Silver layer
22. Gold layer
23. Curated layer
24. Semantic layer
25. Sandbox layer
26. Archive layer
27. Cloud Storage architecture
28. BigLake architecture
29. BigQuery architecture
30. Bigtable use cases
31. Spanner use cases
32. AlloyDB use cases
33. Open-data formats
34. Iceberg considerations
35. Parquet considerations
36. Avro considerations
37. Schema evolution
38. Partitioning
39. Clustering
40. Data lifecycle

41. Retention
42. Archival
43. Legal hold
44. Privacy deletion
45. Metadata architecture
46. Knowledge Catalog
47. Dataplex capabilities
48. Technical metadata
49. Business metadata
50. Business glossary
51. Data lineage
52. Data profiling
53. Data-quality architecture
54. Data contracts
55. Schema registry
56. Data ownership
57. Data stewardship
58. Data certification
59. Data products
60. Data marketplace
61. Analytics Hub
62. Enterprise-data access
63. Data classification
64. Row-level security
65. Column-level security
66. Dynamic data masking
67. Policy tags
68. Authorized views
69. Encryption
70. Data-platform SLOs
71. Data-platform monitoring
72. Cost model
73. Production deployment model

Include:

- End-to-end platform diagram
- Batch-ingestion diagram
- CDC diagram
- Streaming diagram
- Data-layer diagram
- Metadata and governance diagram
- Data-product lifecycle diagram
- Source-to-target mapping
- Data-layer standards
- BigQuery design standards
- Sample dataset and table definitions

# VOLUME 4: Data Engineering Pipelines

Target length: approximately 60 pages.

Produce:

1. Data-engineering strategy
2. Pipeline architecture
3. Pipeline taxonomy
4. Batch processing
5. Micro-batch processing
6. Streaming processing
7. Event-time processing
8. Windowing
9. Stateful processing
10. Deduplication
11. Idempotency
12. Late-arriving data
13. Replay
14. Backfill
15. Reprocessing
16. Data reconciliation
17. ETL
18. ELT
19. Dataflow
20. Apache Beam
21. Dataproc
22. Serverless for Apache Spark
23. Spark
24. PySpark
25. Dataform
26. BigQuery SQL
27. Cloud Composer
28. Apache Airflow
29. Workflows
30. Cloud Run
31. Pub/Sub
32. Datastream
33. BigQuery Storage Write API
34. Pipeline orchestration
35. Dependency management
36. Pipeline metadata
37. Pipeline templates
38. Reusable components
39. Standard connectors
40. Error handling

41. Retry patterns
42. Dead-letter handling
43. Quarantine patterns
44. Checkpointing
45. Schema validation
46. Data-quality validation
47. Pipeline observability
48. Pipeline SLOs
49. Performance tuning
50. Cost optimization
51. Development standards
52. Testing standards
53. Deployment standards
54. Support standards

Generate production-ready examples for:

- Apache Beam
- Dataflow
- Spark
- PySpark
- Dataform SQLX
- BigQuery SQL
- Airflow DAGs
- Pub/Sub publishers and subscribers
- Datastream-to-BigQuery patterns
- Python
- Shell
- YAML
- Terraform

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## VOLUME 5: Enterprise Data Mesh

Target length: approximately 60 pages.

Produce:

1. Data-mesh vision
2. Data-mesh principles
3. Domain architecture
4. Domain decomposition
5. Domain ownership
6. Data-product ownership
7. Central-platform responsibilities
8. Domain-team responsibilities

9. Self-service data platform
10. Federated computational governance
11. Data-product lifecycle
12. Data contracts
13. Data-product interfaces
14. Data-product SLOs
15. Data-product quality
16. Data-product security
17. Data-product versioning
18. Data-product observability
19. Data-product cost allocation
20. Data marketplace
21. Data discovery
22. Analytics Hub
23. Access-request workflows
24. Cross-domain sharing
25. Data-product certification
26. Data-product deprecation
27. Governance council
28. Domain data owners
29. Data stewards
30. Platform product team
31. Domain platform patterns
32. Domain project structure
33. Domain dataset structure
34. Data-mesh IAM model
35. Data-mesh network model
36. Data-mesh governance model
37. Data-mesh operating model
38. Data-mesh maturity model

Develop detailed example domains for:

- Healthcare
- Finance
- Human Resources
- Sales
- Marketing
- Supply Chain

For every domain, include:

- Domain purpose
- Data sources
- Domain owner
- Data steward
- Key data products
- Product consumers

- Data contracts
  - Security classification
  - Quality rules
  - SLOs
  - Access model
  - Project structure
  - Dataset structure
  - Cost-allocation model
- 

## VOLUME 6: Analytics Platform

Target length: approximately 35 pages.

Produce:

1. Analytics strategy
2. Analytics user personas
3. Self-service analytics
4. Governed analytics
5. Looker architecture
6. LookML
7. Semantic modeling
8. Metrics layer
9. KPI governance
10. Certified measures
11. Looker Studio
12. Connected Sheets
13. BigQuery Studio
14. SQL workspaces
15. Notebooks
16. BI Engine
17. Dashboard standards
18. Report standards
19. Semantic-layer standards
20. Row-level security
21. Column-level security
22. Embedded analytics
23. Operational analytics
24. Regulatory reporting
25. Executive dashboards
26. Analytics SLOs
27. Query-performance standards
28. Caching
29. Concurrency
30. Cost optimization

31. Analytics governance
32. Analytics testing
33. Analytics deployment

Include representative:

- Executive dashboard architecture
  - Semantic-model structure
  - LookML examples
  - BigQuery SQL examples
  - KPI-definition templates
  - Analytics-access matrix
- 

## **VOLUME 7: AI, GenAI, RAG and LLMOps**

Target length: approximately 70 pages.

Produce:

1. Enterprise AI strategy
2. AI-ready data platform
3. Vertex AI architecture
4. Gemini integration
5. Vertex AI Agent Builder
6. Enterprise Search
7. Vector Search
8. Embeddings
9. Feature engineering
10. Vertex AI Feature Store
11. Model Registry
12. Model training
13. Model deployment
14. Batch prediction
15. Online prediction
16. Model endpoints
17. Model pipelines
18. Vertex AI Pipelines
19. ML metadata
20. Experiment tracking
21. Model monitoring
22. Model drift
23. Feature drift
24. Bias monitoring
25. Prompt engineering
26. Prompt management

27. Prompt versioning
28. Prompt evaluation
29. LLMOps
30. MLOps
31. Retrieval-augmented generation
32. Enterprise RAG architecture
33. Document ingestion
34. Chunking
35. Embedding generation
36. Vector indexing
37. Retrieval
38. Reranking
39. Grounding
40. Citation generation
41. Guardrails
42. Content safety
43. Model access controls
44. AI data security
45. Sensitive-data protection
46. Prompt-injection protection
47. Data-exfiltration prevention
48. Responsible AI
49. Human oversight
50. AI audit
51. Model cards
52. Document AI
53. Speech-to-Text
54. Text-to-Speech
55. Vision AI
56. Multimodal AI
57. Agent architecture
58. Multi-agent architecture
59. Tool-use security
60. AI observability
61. Token-cost management
62. AI FinOps
63. AI SLOs
64. AI disaster recovery
65. AI governance
66. AI operating model

Include:

- Enterprise RAG diagram
- LLMOps lifecycle diagram
- MLOps architecture
- Agent architecture
- AI security architecture



- Model-governance workflow
  - Vertex AI pipeline examples
  - RAG Python examples
  - Prompt templates
  - Evaluation examples
  - Terraform examples
- 

## VOLUME 8: Security and Zero Trust

Target length: approximately 60 pages.

Produce:

1. Enterprise security strategy
2. Shared-responsibility model
3. Zero Trust architecture
4. Identity perimeter
5. Network perimeter
6. Data perimeter
7. Workload perimeter
8. IAM architecture
9. Human identities
10. Workload identities
11. Service accounts
12. Privileged access
13. Just-in-time access
14. Break-glass access
15. Separation of duties
16. Organization policies
17. VPC Service Controls
18. Access Context Manager
19. Private Service Connect
20. Cloud Armor
21. Cloud Firewall
22. Security Command Center
23. Event Threat Detection
24. Container Threat Detection
25. Secret Manager
26. Cloud KMS
27. Cloud HSM
28. CMEK
29. External Key Manager
30. Key rotation
31. Certificate management
32. Sensitive Data Protection

33. Data classification
34. Tokenization
35. Masking
36. Pseudonymization
37. Audit logging
38. Access Transparency
39. Access Approval
40. Supply-chain security
41. Binary Authorization
42. Artifact Analysis
43. Vulnerability scanning
44. Security monitoring
45. SIEM integration
46. SOAR integration
47. Threat modeling
48. STRIDE
49. Attack-path analysis
50. Incident response
51. Regulatory control mapping
52. Security testing
53. Penetration testing
54. Security assurance
55. Security architecture review

Provide control mappings for:

- NIST
- HIPAA
- GDPR
- PCI DSS
- SOC 2
- ISO 27001

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## **VOLUME 9: DevSecOps and CI/CD**

Target length: approximately 50 pages.

Produce:

1. DevSecOps strategy
2. Source-control strategy
3. GitHub architecture
4. Repository structure
5. Monorepo versus multi-repo
6. Branching strategy

7. Versioning strategy
8. Release strategy
9. Cloud Build
10. GitHub Actions
11. Artifact Registry
12. Cloud Deploy
13. Terraform pipelines
14. Application pipelines
15. Dataform pipelines
16. Dataflow pipelines
17. Airflow deployment
18. BigQuery deployment
19. Schema deployment
20. Policy-as-Code
21. Security scanning
22. Secret scanning
23. Dependency scanning
24. Container scanning
25. Terraform validation
26. Unit testing
27. Integration testing
28. Data testing
29. Quality testing
30. Performance testing
31. Security testing
32. Environment promotion
33. Development
34. Integration
35. Test
36. Staging
37. Production
38. Approval workflows
39. Change management
40. Rollback
41. Blue-green deployment
42. Canary deployment
43. Immutable artifacts
44. Software supply-chain security
45. SLSA alignment
46. Auditability
47. Deployment monitoring
48. Release metrics

Generate production-ready:

- Cloud Build YAML
- GitHub Actions workflows
- Terraform pipelines

- Policy validation
  - Test stages
  - Promotion stages
  - Rollback procedures
- 

## VOLUME 10: FinOps and Cost Governance

Target length: approximately 40 pages.

Produce:

1. Enterprise FinOps strategy
2. FinOps operating model
3. Cost ownership
4. Billing-account structure
5. Cost-center mapping
6. Resource labeling
7. Resource tagging
8. Showback
9. Chargeback
10. Budget management
11. Budget alerts
12. Forecasting
13. Cost anomaly detection
14. Unit economics
15. BigQuery cost model
16. BigQuery editions
17. Reservations
18. Slot allocation
19. Slot autoscaling
20. Workload management
21. Query optimization
22. Partition optimization
23. Clustering optimization
24. Materialized views
25. Storage optimization
26. Cloud Storage classes
27. Lifecycle policies
28. Dataflow optimization
29. Dataproc optimization
30. Pub/Sub optimization
31. Network-cost optimization
32. Logging-cost optimization
33. Monitoring-cost optimization
34. Vertex AI cost optimization

35. Gemini token optimization
36. Idle-resource management
37. Commitment planning
38. Cost dashboards
39. FinOps KPIs
40. Monthly cost reviews
41. Quarterly optimization reviews

Do not invent current product prices.

Create cost models based on measurable variables that can be entered into the current Google Cloud Pricing Calculator.

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## VOLUME 11: Monitoring, SRE and Operations

Target length: approximately 50 pages.

Produce:

1. Operations strategy
2. SRE operating model
3. Service ownership
4. Platform ownership
5. Data-product ownership
6. SLIs
7. SLOs
8. SLAs
9. Error budgets
10. Observability architecture
11. Cloud Monitoring
12. Cloud Logging
13. Managed Service for Prometheus
14. OpenTelemetry
15. Cloud Trace
16. Error Reporting
17. Application metrics
18. Pipeline metrics
19. Data metrics
20. Data-freshness metrics
21. Data-quality metrics
22. BigQuery metrics
23. Dataflow metrics
24. Pub/Sub metrics
25. Vertex AI metrics
26. Security metrics

27. FinOps metrics
28. Dashboard standards
29. Alerting standards
30. Alert severity
31. Alert routing
32. On-call model
33. Runbooks
34. Playbooks
35. Incident management
36. Problem management
37. Change management
38. Capacity planning
39. Performance management
40. Operational readiness review
41. Production readiness review
42. Post-incident review
43. Chaos engineering
44. Resilience testing
45. Service health
46. Support escalation
47. Vendor escalation

Provide:

- SLI and SLO catalog
- Alert matrix
- Runbook templates
- Incident-response workflow
- Operations dashboards
- Production-readiness checklist

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## **VOLUME 12: Disaster Recovery and Business Continuity**

Target length: approximately 35 pages.

Produce:

1. Business-impact analysis
2. Service-tier classification
3. RTO
4. RPO
5. Regional-failure scenarios
6. Zone-failure scenarios

7. Dependency-failure scenarios
8. Data-corruption scenarios
9. Cyber-recovery scenarios
10. Backup strategy
11. Restore strategy
12. BigQuery recovery
13. Cloud Storage recovery
14. Pub/Sub replay
15. Dataflow recovery
16. Datastream recovery
17. Composer recovery
18. Looker recovery
19. Vertex AI recovery
20. Metadata recovery
21. Terraform-state recovery
22. Source-code recovery
23. Configuration recovery
24. Cross-region replication
25. Regional failover
26. DNS failover
27. Recovery automation
28. DR runbooks
29. DR testing
30. Tabletop exercises
31. Failover exercises
32. Failback
33. Business continuity
34. Crisis communication
35. DR governance

Include workload-level RPO/RTO matrices and service-specific recovery patterns.

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## **VOLUME 13: Terraform and Infrastructure as Code**

Target length: approximately 70 pages.

Produce:

1. Infrastructure-as-Code strategy
2. Terraform architecture
3. Repository design
4. Module design
5. Root modules
6. Reusable modules
7. Environment configuration

8. Variable strategy
9. State management
10. Remote state
11. State locking
12. State security
13. Provider management
14. Version pinning
15. Organization deployment
16. Folder deployment
17. Project factory
18. Shared VPC deployment
19. IAM deployment
20. KMS deployment
21. VPC Service Controls deployment
22. Logging deployment
23. Monitoring deployment
24. Cloud Storage deployment
25. BigQuery deployment
26. Pub/Sub deployment
27. Dataflow deployment
28. Dataplex deployment
29. Composer deployment
30. Vertex AI deployment
31. Artifact Registry deployment
32. Cloud Build deployment
33. Policy-as-Code
34. Terraform testing
35. Static analysis
36. Security scanning
37. Plan review
38. Approval
39. Apply
40. Drift detection
41. Import strategy
42. Migration strategy
43. Rollback
44. Disaster recovery
45. Module documentation
46. Module versioning
47. Module publishing
48. Golden modules
49. Landing-zone modules
50. Domain modules

Generate production-ready Terraform examples and repository structures.

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# VOLUME 14: Industry Reference Implementations

Target length: approximately 100 pages.

Produce complete reference implementations for:

1. Healthcare
2. Banking and financial services
3. Retail
4. Telecommunications

For each industry, include:

- Business context
  - Business capabilities
  - Regulatory requirements
  - Data sources
  - Reference architecture
  - Landing-zone considerations
  - Data ingestion
  - Data layers
  - Data products
  - Data mesh
  - Analytics
  - AI and GenAI
  - Security
  - Networking
  - DevSecOps
  - SRE
  - Disaster recovery
  - FinOps
  - Implementation roadmap
  - Architecture Decision Records
  - Risk register
  - Terraform examples
  - Pipeline examples
  - BigQuery examples
  - AI examples
  - Professional diagrams
-

## 4. Required Analysis for Every Google Cloud Service

For every Google Cloud service selected, provide a structured service assessment containing:

1. Service name
2. Purpose
3. Enterprise architecture role
4. Business capabilities supported
5. Technical capabilities
6. Benefits
7. Limitations
8. Service quotas
9. Regional availability considerations
10. High-availability characteristics
11. Disaster-recovery considerations
12. Security model
13. IAM requirements
14. Network requirements
15. VPC Service Controls considerations
16. Encryption capabilities
17. CMEK support
18. Logging
19. Monitoring
20. Alerting
21. Cost drivers
22. Cost-optimization techniques
23. Operational responsibilities
24. Integration points
25. Alternatives
26. Selection criteria
27. Best practices
28. Reference architecture
29. Sample configuration
30. Common pitfalls
31. Anti-patterns
32. Production-readiness checklist

Do not include a service simply because it exists.

Every selected service must map to one or more documented business, security, data, AI or operational requirements.

## 5. Required Architecture Decision Record Format

Every major architecture decision must include:

- ADR ID
- Decision title
- Status
- Date
- Architecture domain
- Problem
- Context
- Decision drivers
- Constraints
- Assumptions
- Alternatives considered
- Evaluation criteria
- Decision
- Rationale
- Benefits
- Tradeoffs
- Positive consequences
- Negative consequences
- Risks
- Mitigations
- Implementation implications
- Cost implications
- Security implications
- Operational implications
- Review triggers
- Review date

At minimum, create ADRs for:

- Google Cloud organization structure
- Folder structure
- Project strategy
- Shared VPC
- Region strategy
- Multi-region strategy
- Private Service Connect
- Hybrid connectivity
- VPC Service Controls
- CMEK
- BigQuery
- Cloud Storage
- BigLake
- Dataflow

- Dataproc
- Dataform
- Cloud Composer
- Pub/Sub
- Datastream
- Batch versus streaming
- ETL versus ELT
- Lakehouse architecture
- Data-mesh architecture
- Data-product ownership
- Governance model
- Looker semantic layer
- Vertex AI
- RAG architecture
- Vector database
- LLMOps
- Terraform
- Cloud Build versus GitHub Actions
- BigQuery reservation strategy
- Disaster-recovery model

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## 6. Required Professional Diagrams

Produce professional diagrams in multiple compatible formats.

For every major architecture, generate:

- Mermaid
- PlantUML
- Graphviz DOT
- SVG-compatible structure
- draw.io-compatible design specification

Where direct draw.io XML generation is practical, generate valid draw.io XML.

Where complete draw.io XML would make the document unreadable, provide:

- Page structure
- Swimlanes
- Containers
- Components
- Connectors
- Labels
- Trust boundaries
- Regions

- Zones
- Legend
- Recommended Google Cloud icons
- draw.io construction instructions

Generate diagrams for:

1. Enterprise conceptual architecture
2. Enterprise logical architecture
3. Enterprise physical architecture
4. Organization and folder hierarchy
5. Landing zone
6. Shared VPC
7. Hybrid connectivity
8. Identity architecture
9. Security architecture
10. VPC Service Controls
11. Enterprise data platform
12. Batch ingestion
13. CDC ingestion
14. Streaming ingestion
15. Data layers
16. BigQuery architecture
17. BigLake architecture
18. Governance architecture
19. Data lineage
20. Data mesh
21. Data-product lifecycle
22. Analytics platform
23. AI platform
24. RAG architecture
25. LLMOps lifecycle
26. DevSecOps
27. CI/CD
28. Terraform deployment
29. Observability
30. Disaster recovery
31. FinOps
32. Industry reference architectures

Diagram quality requirements:

- Use Google Cloud architecture conventions.
- Use official Google Cloud service names.
- Group services by architectural plane.
- Show regions and zones.
- Show trust boundaries.
- Show service perimeters.

- Show environment boundaries.
  - Show data-flow direction.
  - Show control flow separately where useful.
  - Include a legend.
  - Avoid unreadable diagrams.
  - Split complex architecture into multiple views.
  - Explain every diagram.
- 

## 7. Required Production-Ready Examples

Generate production-quality examples where relevant.

### Infrastructure

- Terraform
- YAML
- JSON
- Shell scripts
- Organization policies
- IAM policies
- Firewall policies
- VPC Service Controls configurations
- Cloud KMS configurations
- Monitoring configurations
- Alerting policies

### CI/CD

- Cloud Build
- GitHub Actions
- Cloud Deploy
- Terraform plans
- Security scanning
- Policy validation
- Test automation
- Environment promotion
- Rollback

### Data Engineering

- Apache Beam
- Dataflow
- Apache Spark
- PySpark
- Airflow DAGs

- Dataform SQLX
- BigQuery SQL
- Pub/Sub
- Datastream patterns
- Python
- Shell

## AI and ML

- Vertex AI Pipelines
- Model registration
- Batch prediction
- Online prediction
- Embedding generation
- Vector Search
- RAG
- Gemini integration
- Prompt templates
- Prompt evaluation
- Model monitoring
- Responsible-AI controls

Every code example must include:

- Purpose
- Prerequisites
- Configuration
- Error handling
- Logging
- Security considerations
- Testing instructions
- Deployment instructions
- Rollback instructions
- Production limitations

Do not label pseudocode as production-ready code.

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## 8. Required Tables and Matrices

Generate, where relevant:

- Business-requirements matrix
- Functional-requirements matrix
- Non-functional-requirements matrix
- Security-requirements matrix
- Compliance-control matrix

- Capability matrix
- Service-mapping matrix
- Technology-selection matrix
- Service-comparison matrix
- Source-to-target mapping
- Data-classification matrix
- Data-layer matrix
- Data-product catalog
- Data-contract matrix
- Governance RACI
- IAM matrix
- Network-flow matrix
- Firewall-rule matrix
- VPC Service Controls matrix
- Encryption matrix
- Key-management matrix
- Data-quality rules
- SLI and SLO catalog
- Alerting matrix
- RPO/RTO matrix
- Cost-driver matrix
- Chargeback matrix
- Risk register
- Architecture Decision Records
- Migration-wave plan
- Implementation roadmap
- Responsibility matrix
- Production-readiness checklist

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## 9. Documentation Quality Standards

The blueprint must:

- Be written in professional enterprise-consulting language.
- Be understandable to executive and technical audiences.
- Separate business, conceptual, logical and physical architecture.
- Explain why each architecture choice was made.
- Avoid generic descriptions.
- Include implementation guidance.
- Include production considerations.
- Include security in every layer.
- Include operations in every layer.
- Include cost considerations in every layer.
- Include failure scenarios.
- Include alternatives and tradeoffs.



- Include common pitfalls.
- Include anti-patterns.
- Include diagrams.
- Include tables.
- Include code examples.
- Include validation checklists.
- Include measurable acceptance criteria.
- Include open decisions.
- Include assumptions.
- Include risks and mitigations.
- Use consistent terminology across all volumes.
- Use consistent naming conventions across all examples.
- Avoid unsupported claims.
- Avoid invented product features.
- Identify items that require validation against current Google Cloud documentation.

The content must be suitable for:

- Enterprise Architecture Review Board approval
- Security Architecture Review Board approval
- Cloud Center of Excellence review
- Data Governance Council approval
- Regulatory and compliance review
- Platform-engineering implementation
- Production-readiness review
- Financial approval

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## 10. Delivery Method

Do not attempt to generate all 800–900 pages in a single response.

Generate the blueprint as a controlled series.

### Phase 1: Series Foundation

First produce:

1. Master table of contents
2. Architecture assumptions
3. Enterprise terminology
4. Naming conventions
5. Architecture principles
6. Global decisions
7. Cross-volume dependency map
8. Deliverable tracker

## Phase 2: Volume Delivery

Generate one complete volume at a time.

For each volume:

1. Show the volume title.
2. Show document control.
3. Show the table of contents.
4. State inherited assumptions.
5. Produce every required section.
6. Produce the required diagrams.
7. Produce the required tables.
8. Produce the required code examples.
9. Produce relevant ADRs.
10. Produce risks and mitigations.
11. Produce implementation guidance.
12. Produce validation criteria.
13. Produce production-readiness checks.
14. Identify dependencies on other volumes.
15. Conclude with decisions and next actions.

## Phase 3: Cross-Volume Validation

After all volumes are complete:

- Validate architecture consistency.
  - Validate service selections.
  - Validate region strategy.
  - Validate security controls.
  - Validate IAM boundaries.
  - Validate network paths.
  - Validate data flows.
  - Validate data-product ownership.
  - Validate compliance controls.
  - Validate RPO and RTO targets.
  - Validate monitoring coverage.
  - Validate FinOps controls.
  - Validate implementation dependencies.
  - Validate naming conventions.
  - Validate code consistency.
  - Produce a final architecture-assurance report.
-

## 11. Estimated Blueprint Series

| Volume | Title                                     | Estimated length        |
|--------|-------------------------------------------|-------------------------|
| 1      | Executive Architecture Guide              | Approximately 40 pages  |
| 2      | Google Cloud Landing Zone                 | Approximately 50 pages  |
| 3      | Enterprise Data Platform                  | Approximately 70 pages  |
| 4      | Data Engineering Pipelines                | Approximately 60 pages  |
| 5      | Enterprise Data Mesh                      | Approximately 60 pages  |
| 6      | Analytics Platform                        | Approximately 35 pages  |
| 7      | AI, GenAI, RAG and LLMOps                 | Approximately 70 pages  |
| 8      | Security and Zero Trust                   | Approximately 60 pages  |
| 9      | DevSecOps and CI/CD                       | Approximately 50 pages  |
| 10     | FinOps and Cost Governance                | Approximately 40 pages  |
| 11     | Monitoring, SRE and Operations            | Approximately 50 pages  |
| 12     | Disaster Recovery and Business Continuity | Approximately 35 pages  |
| 13     | Terraform and Infrastructure as Code      | Approximately 70 pages  |
| 14     | Industry Reference Implementations        | Approximately 100 pages |

Estimated total: approximately 800–900 pages of reusable enterprise architecture guidance.

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## 12. Final Execution Instruction

Begin by producing the **Enterprise Blueprint Series Foundation**, containing:

1. Master table of contents for all 14 volumes
2. Fortune 100 architecture assumptions
3. Enterprise capability model
4. Enterprise conceptual reference architecture
5. Cross-volume dependency map
6. Global naming conventions
7. Global security classifications
8. Global environment model
9. Global region and resilience strategy
10. Global architecture principles
11. Global technology-selection principles

12. Global Architecture Decision Record index
13. Deliverable-production plan
14. Quality-assurance checklist

After completing the foundation, produce **Volume 1: Executive Architecture Guide**.

Do not ask the user to complete placeholders.

Make reasonable assumptions and clearly document them.

Do not reduce the architecture to a high-level summary.

Produce professional consulting-quality architecture documentation, diagrams, matrices and implementation examples.